
Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

Probability & Statistics

BCA — Semester 3 — 2019 — Answer Sheet

Subject Code: CAST202

Group A

Attempt ALL the questions. $[10 \times 1 = 10]$

1. i) How many types of data on the basis of sources of data collection?

- a) 1 b) 2 c) 3 d) 4

ii) Which is more appropriate central tendency to find the average of profit?

- a) Arithmetic mean b) Median c) Mode d) All

iii) What is the range of Correlation?

- a) 0 to ∞ b) $-\infty$ to ∞ c) -1 to 1 d) 0 to 1

iv) If $r=0.2$ then coefficient of determination implies that

- a) 20% of total variation in dependent variable has been explained by independent variable.
b) 40% of total variation...
c) 2% of total variation...
d) 4% of total variation...

v) What is the minimum value of Probability?

- a) 1 b) 100 c) 0 d) None of above

vi) In case of Normal distribution

- a) Mean > Median b) Mean = Median c) Mean $\leq$ Median d) Mean $\geq$ Median

vii) The regression line of X on Y and Y on X are intersect at the point

- a) $(\mu, 0)$ b) (a, b) c) (X, Y) d) (X, Y)

viii) In case of systematic sampling

- a) sample mean is biased estimator population mean.
b) sample mean is unbiased estimator population mean.
c) sample mean can't estimate population mean.
d) sample mean may equal to population mean.

ix) Mean of Chi-Square distribution with n degrees of freedom is

- a) 1 b) 0 c) 2n d) n

x) How do you obtain degree of freedom in one-way ANOVA?

- a) $(k, n-1)$ b) $(k, n-k)$ c) $(k-1, n-1)$ d) $(k-1, n-k)$

[10 × 1 = 10]

Answer: Answers:

i) **b) 2** — On the basis of sources, data is classified into two types: Primary data (collected directly by the investigator) and Secondary data (collected by someone else and used by another).

ii) **c) Mode** — Mode is the most appropriate measure of central tendency for finding the average profit because profit data often has extreme values (outliers) and mode represents the most frequently occurring value.

iii) **c) -1 to 1** — The correlation coefficient (r) always lies between -1 and +1 inclusive. -1 indicates perfect negative correlation, +1 indicates perfect positive correlation, and 0 indicates no correlation.

iv) **d) 4%** — Coefficient of determination = $r^2 = (0.2)^2 = 0.04 = 4\%$. It means 4% of the total variation in the dependent variable is explained by the independent variable.

v) c) 0 — The minimum value of probability is 0 (impossible event) and the maximum value is 1 (certain event). Probability always lies between 0 and 1.

vi) b) **Mean = Median** — In a normal distribution, the mean, median, and mode are all equal and located at the center of the distribution. The curve is perfectly symmetric.

vii) d) $(\bar{X}, \bar{Y})$ — The two regression lines (X on Y and Y on X) always intersect at the point of means $(\bar{X}, \bar{Y})$, which is the mean of X and mean of Y.

viii) b) **sample mean is unbiased estimator population mean.** — In systematic sampling, the sample mean is an unbiased estimator of the population mean, provided the population is randomly ordered.

ix) d) n — For a Chi-Square distribution with n degrees of freedom, the mean is n and the variance is 2n.

x) d) **(k-1, n-k)** — In one-way ANOVA, degrees of freedom for treatment (between groups) = k-1, and degrees of freedom for error (within groups) = n-k, where k = number of groups and n = total number of observations.

Group B

Attempt any SIX questions. $[6 \times 5 = 30]$

2. Describe scope and limitation of Statistics. [5]

[5]

Answer: Scope of Statistics:

- 1. Business and Economics:** Statistics is extensively used in business for demand forecasting, market research, quality control, pricing decisions, and sales analysis. In economics, it helps in calculating GDP, inflation rates, unemployment rates, and economic planning.
- 2. Government and Administration:** Governments use statistics for census, budget preparation, policy formulation, public health planning, and resource allocation.
- 3. Natural and Social Sciences:** In biology (biostatistics), medicine (clinical trials), psychology (behavioral studies), sociology (surveys), and education (test scoring).
- 4. Research:** Statistical methods are essential for designing experiments, analyzing data, testing hypotheses, and drawing valid conclusions in all fields of research.
- 5. Banking and Finance:** Risk assessment, portfolio management, credit scoring, insurance premium calculation, and stock market analysis.

Limitations of Statistics:

- 1. Deals with aggregates only:** Statistics cannot study individual cases. It studies groups, not single observations.
- 2. Homogeneous data required:** Statistical methods are applicable only to homogeneous (similar) data. Heterogeneous data must be separated before analysis.
- 3. Results are true on average:** Statistical laws are not exact like mathematical laws. They are true on average and under certain conditions.
- 4. Can be misused:** Statistics can be manipulated to support false conclusions. Misleading graphs, selective data, and biased sampling can distort reality.
- 5. Does not study qualitative phenomena:** Statistics primarily deals with numerical data. Qualitative aspects like beauty, honesty, and intelligence cannot be directly measured statistically.

3. Determine average wages from following data:

Wages 25-30 30-35 35-40 40-45 45-50 50-55 55-60 60-65 65-70
No. of Worker 10 13 18 21 24 28 32 33 42

[5]

Answer: Solution: We calculate the mean using the assumed mean method.

Wages f Mid (x) $d=(x-47.5)/5$ fd
25-30 10 27.5 -4 -40
30-35 13 32.5 -3 -39
35-40 18 37.5 -2 -36
40-45 21 42.5 -1 -21
45-50 24 47.5 0 0
50-55 28 52.5 1 28
55-60 32 57.5 2 64
60-65 33 62.5 3 99
65-70 42 67.5 4 168
Total $N=153$ $\Sigma fd=-3$

Assumed mean $A = 47.5$, class width $h = 5$

Mean $= A + (\Sigma fd/N) \times h = 47.5 + (-3/153) \times 5 = 47.5 - 0.098 = 47.40$

Average wages = Rs. 47.40

4. Calculate Karl Pearson's correlation coefficient from the following data:

Sales 4341363450 Expenses 1022131917

[5]

Answer: Solution:

Sales (X) Expenses (Y) $x = X - \bar{X}$ $y = Y - \bar{Y}$ x^2 y^2 xy 43103.8-6.214.4438.44-23.56 41221.85.83.2433.6410.44
3613-3.2-3.210.2410.2410.24 3419-5.22.827.047.84-14.56 501710.80.8116.640.648.64
 $\Sigma X = 204$ $\Sigma Y = 81$ $\Sigma x^2 = 171.6$ $\Sigma y^2 = 90.8$ $\Sigma xy = -8.8$
 $\bar{X} = 204/5 = 40.8$, $\bar{Y} = 81/5 = 16.2$

$$r = \Sigma xy / \sqrt{(\Sigma x^2 \times \Sigma y^2)} = -8.8 / \sqrt{(171.6 \times 90.8)} = -8.8 / \sqrt{15581.28} = -8.8 / 124.825 = -0.0705$$

Karl Pearson's $r \approx -0.07$ (very weak negative correlation)

This indicates almost no linear relationship between Sales and Expenses.

5. Estimate the marks in JAVA when the marks in Statistics is 65 by using following data:

Marks in Statistics 5758595960616264 Marks in JAVA 7778757882827981

[5]

Answer: Solution: We need to find the regression equation of Y (JAVA) on X (Statistics).

$X \ Y \ x = X - 60 \ y = Y - 79 \ x^2 \ y^2 \ xy$ 5777-3-2946 5878-2-1412 5975-1-41164 5978-1-1111 608203090 618213193
627920400 6481421648 $\Sigma X = 480$ $\Sigma Y = 632$ $\Sigma x^2 = 36$ $\Sigma y^2 = 44$ $\Sigma xy = 24$
 $\bar{X} = 480/8 = 60$, $\bar{Y} = 632/8 = 79$

$$\text{Regression coefficient } b_{yx} = \Sigma xy / \Sigma x^2 = 24/36 = 0.667$$

Regression equation: $Y - \bar{Y} = b_{yx}(X - \bar{X})$

$$Y - 79 = 0.667(X - 60)$$

$$Y = 0.667X - 40 + 79$$

$$Y = 0.667X + 39$$

When $X = 65$:

$$Y = 0.667(65) + 39 = 43.355 + 39 = 82.355 \approx 82$$

Estimated marks in JAVA when Statistics = 65 is approximately 82 marks.

6. Fit Binomial Distribution from the following data where $p = 0.5$:

No. of heads 0 1 2 3 4 **Frequency** 28 62 46 20 4

[5]

Answer: Solution:

Given: $p = 0.5$, $q = 1 - p = 0.5$

$N = \text{Total frequency} = 28 + 62 + 46 + 20 + 4 = 160$

$n = 4$ (maximum number of heads)

The binomial probabilities are calculated using $P(X=r) = {}^nC_r p^r q^{n-r}$

$$P(X=0) = {}^nC_0 (0.5)^0 (0.5)^4 = 1 \times 1 \times 0.0625 = 0.0625$$

$$\text{Expected frequency} = 160 \times 0.0625 = \mathbf{10}$$

$$P(X=1) = {}^nC_1 (0.5)^1 (0.5)^3 = 4 \times 0.5 \times 0.125 = 0.25$$

$$\text{Expected frequency} = 160 \times 0.25 = \mathbf{40}$$

$$P(X=2) = {}^nC_2 (0.5)^2 (0.5)^2 = 6 \times 0.25 \times 0.25 = 0.375$$

$$\text{Expected frequency} = 160 \times 0.375 = \mathbf{60}$$

$$P(X=3) = {}^nC_3 (0.5)^3 (0.5)^1 = 4 \times 0.125 \times 0.5 = 0.25$$

$$\text{Expected frequency} = 160 \times 0.25 = \mathbf{40}$$

$$P(X=4) = {}^nC_4 (0.5)^4 (0.5)^0 = 1 \times 0.0625 \times 1 = 0.0625$$

$$\text{Expected frequency} = 160 \times 0.0625 = \mathbf{10}$$

No. of heads (x)	0	1	2	3	4
Total Observed (f)	28	62	46	20	4
Expected	10	40	60	40	10

The fitted binomial distribution shows the expected frequencies under the theoretical binomial model with $p=0.5$.

7. How do you determine sample size in sampling? Explain briefly. [5]

[5]

Answer: Determination of Sample Size:

Sample size is the number of observations or individuals included in a sample. It is crucial for obtaining reliable and valid results. The sample size depends on several factors:

1. Population Size (N): For very large populations, the sample size formula becomes independent of N. For small populations, finite population correction is applied.

2. Margin of Error (E): The maximum acceptable difference between sample statistic and population parameter. Smaller margin requires larger sample size.

3. Confidence Level: Common levels are 90%, 95%, and 99%. Higher confidence requires larger sample size. Z-values: 90% → 1.645, 95% → 1.96, 99% → 2.576.

4. Population Standard Deviation (σ): Higher variability requires larger sample size.

Formulas:

For estimating mean: $n = (Z^2 \times \sigma^2) / E^2$

For estimating proportion: $n = (Z^2 \times p \times q) / E^2$

Where:

- Z = critical value from standard normal distribution
- σ = population standard deviation (or sample estimate)
- E = margin of error
- p = estimated proportion, q = 1-p

Example: To estimate average income with 95% confidence, margin of error = Rs. 100, and σ = Rs. 500:

$$n = (1.96^2 \times 500^2) / 100^2 = (3.8416 \times 250000) / 10000 = 960400 / 10000 = 96.04 \approx 97$$

Practical Considerations:

- Budget constraints may limit sample size
- Response rate should be considered (increase sample size accordingly)
- For stratified sampling, allocate samples proportionally to strata sizes

8. Write short notes on simple random sampling. [5]

[5]

Answer: Simple Random Sampling (SRS):

Simple Random Sampling is a probability sampling technique where every member of the population has an equal and independent chance of being selected in the sample. It is the most basic and fundamental sampling method.

Methods of Selection:

1. Lottery Method: Each member of the population is assigned a number, numbers are written on identical slips, mixed thoroughly, and the required number of slips are drawn blindly.

2. Random Number Table: Using published tables of random digits (like Tippett's table, Fisher-Yates table) to select sample members.

3. Computer-Generated Random Numbers: Using software functions like RAND() in Excel or random module in Python to generate random numbers.

Advantages:

- Free from personal bias and subjectivity
- Easy to understand and implement
- Every unit has equal chance of selection
- Sample is representative if population is homogeneous
- Statistical theory is well-developed for inference

Disadvantages:

- Requires complete list of population (sampling frame)
- May not represent subgroups well if population is heterogeneous
- Can be expensive and time-consuming for large, dispersed populations
- Sample may be geographically scattered, increasing survey costs

When to use: SRS is most suitable when the population is small, homogeneous, and a complete sampling frame is available.

Group C

Attempt any TWO questions. [$2 \times 10 = 20$]

9. Student's age in the regular daytime BCA program and the morning time BCA program of a campus are described by two samples. If the homogeneity in age of the class is positive factor in learning make suggestion, with reason, which of two groups will be easier to teach?

Regular BCA program Morning BCA program AgeNo. of Students AgeNo. of Students
 2392710292318285305221029430128621433525113452663511273362249324 Total60 Total60

[10]

Answer: Solution: We compare homogeneity using Coefficient of Variation (CV). Lower CV means more homogeneous.

For Regular BCA Program:

Age (X) $\sum X^2$ 214844411764 22102204844840 2392075294761 2492165765184 25112756256875
 2661566764056 273817292187 2851407843920 292588411682 30130900900 **Total60146736169**

Mean = $1467/60 = 24.45$

Variance = $(36169/60) - (24.45)^2 = 602.817 - 597.803 = 5.014$

SD = $\sqrt{5.014} = 2.24$

CV = $(2.24/24.45) \times 100 = 9.16\%$

For Morning BCA Program:

Age (Y) $\sum Y^2$ 27102707297290 2861687844704 2941168413364 3051509004500 3182489617688
 32412810244096 33516510895445 34517011565780 3511385122513475 3627212962592
Total60187258934

Mean = $1872/60 = 31.2$

Variance = $(58934/60) - (31.2)^2 = 982.233 - 973.44 = 8.793$

SD = $\sqrt{8.793} = 2.965$

CV = $(2.965/31.2) \times 100 = 9.50\%$

Conclusion:

The Regular BCA program has CV = 9.16%, while Morning BCA has CV = 9.50%. Since the Regular BCA program has a lower coefficient of variation, it is **more homogeneous** in age.

Suggestion: The **Regular BCA program** will be easier to teach because the students' ages are more uniform (less variation). Homogeneous age groups tend to have similar learning paces, prior knowledge levels, and cognitive development, making it easier for instructors to design and deliver lessons effectively.

10. Given a normal distribution with mean 200 and s.d. 20, find the probability that

i) $P(X > 180)$ ii) $P(X < 220)$

iii) $P(160 < X < 240)$

v) $P(X < 180 \text{ or } X > 220)$ vi) 10% of the values are less than what values of X?

[10]

Answer: Solution: Given: $\mu = 200$, $\sigma = 20$

We use $Z = (X - \mu) / \sigma$

i) $P(X > 180)$:

$$Z = (180 - 200) / 20 = -1$$

$$P(X > 180) = P(Z > -1) = P(Z < 1) = \mathbf{0.8413 \text{ or } 84.13\%}$$

ii) $P(X < 220)$:

$$Z = (220 - 200) / 20 = 1$$

$$P(X < 220) = P(Z < 1) = \mathbf{0.8413 \text{ or } 84.13\%}$$

iii) $P(160 < X < 240)$:

$$Z_{\blacksquare} = (160 - 200) / 20 = -2$$

$$Z_{\blacksquare} = (240 - 200) / 20 = 2$$

$$P(-2 < Z < 2) = P(Z < 2) - P(Z < -2) = 0.9772 - 0.0228 = \mathbf{0.9544 \text{ or } 95.44\%}$$

iv) $P(X > 220)$:

$$Z = (220 - 200) / 20 = 1$$

$$P(X > 220) = P(Z > 1) = 1 - P(Z < 1) = 1 - 0.8413 = \mathbf{0.1587 \text{ or } 15.87\%}$$

v) $P(X < 180 \text{ or } X > 220)$:

$$P(X < 180) = P(Z < -1) = 0.1587$$

$$P(X > 220) = 0.1587 \text{ (from iv)}$$

$$\text{Since these are mutually exclusive: } P(X < 180 \text{ or } X > 220) = 0.1587 + 0.1587 = \mathbf{0.3174 \text{ or } 31.74\%}$$

$$\text{Alternatively: } 1 - P(180 < X < 220) = 1 - [P(Z < 1) - P(Z < -1)] = 1 - [0.8413 - 0.1587] = 1 - 0.6826 = 0.3174$$

vi) Find X such that $P(X < x) = 10\% = 0.10$:

From Z-table, $P(Z < -1.28) \approx 0.10$

$$Z = (X - 200) / 20 = -1.28$$

$$X = 200 + (-1.28 \times 20) = 200 - 25.6 = \mathbf{174.4}$$

10% of the values are less than 174.4.

11. The labor productivity indexes of Nepal are recorded as below:

Sector	Year 2015	Year 2016	Year 2017	Row Total (R _i)
Agriculture	100	125	138	363
Manufacturing	100	60	53	213
Community & Social service	100	89	80	269
Col Total (C _j)	300	274	271	G=845

Does the labor productivity index vary due to the;
i) difference in the sector ii) difference in the time period?

[10]

Answer: Solution: This is a Two-Way ANOVA problem.

Step 1: Calculate row totals, column totals, and grand total

Sector 2015 2016 2017 Row Total (R_i) Agriculture 100 125 138 363 Manufacturing 100 60 53 213 Community & Social 100 89 80 269 Col Total (C_j) 300 274 271 G=845
N = 9, k = 3 sectors, h = 3 years

Step 2: Calculate Correction Factor (CF)

$$CF = G^2/N = 845^2/9 = 714025/9 = 79336.11$$

Step 3: Calculate Total Sum of Squares (TSS)

$$\begin{aligned} TSS &= \sum X^2 - CF = (100^2 + 125^2 + 138^2 + 100^2 + 60^2 + 53^2 + 100^2 + 89^2 + 80^2) - 79336.11 \\ &= (10000 + 15625 + 19044 + 10000 + 3600 + 2809 + 10000 + 7921 + 6400) - 79336.11 \\ &= 85399 - 79336.11 = 6062.89 \end{aligned}$$

Step 4: Calculate Sum of Squares between Sectors (SSS)

$$\begin{aligned} SSS &= (\sum R_i^2/h) - CF = (363^2 + 213^2 + 269^2)/3 - 79336.11 \\ &= (131769 + 45369 + 72361)/3 - 79336.11 \\ &= 249499/3 - 79336.11 = 83166.33 - 79336.11 = 3830.22 \end{aligned}$$

Step 5: Calculate Sum of Squares between Years (SSY)

$$\begin{aligned} SSY &= (\sum C_j^2/k) - CF = (300^2 + 274^2 + 271^2)/3 - 79336.11 \\ &= (90000 + 75076 + 73441)/3 - 79336.11 \\ &= 238517/3 - 79336.11 = 79505.67 - 79336.11 = 169.56 \end{aligned}$$

Step 6: Calculate Error Sum of Squares (SSE)

$$SSE = TSS - SSS - SSY = 6062.89 - 3830.22 - 169.56 = 2063.11$$

Step 7: ANOVA Table

Source	SS	df	MSS
Sectors	3830.22	19	15.11
Years	169.56	2	84.78
Error	2063.11	45	15.78
Total	6062.89	68	

Step 8: Compare with F-critical

F-critical (2,4) at 5% = 6.94

For Sectors: F = 1.86 < 6.94 → **Not significant**

For Years: F = 0.08 < 6.94 → **Not significant**

Conclusion:

- i) **No**, the labor productivity index does not vary significantly due to difference in sectors (F = 1.86 < 6.94).
- ii) **No**, the labor productivity index does not vary significantly due to difference in time periods (F = 0.08 < 6.94).

Note: The small sample size and limited data may contribute to the non-significant results.

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.